

Noninjurious C6C7 Shear Supplemental Data – Guide to fixturing

The following pages contain critical information on how the fixturing was set up and used in these experiments. A detailed Bill of Materials (BOM) is included, as well as a diagram of parts and assembly details. The repository includes models of all parts in Solidworks and .STEP file formats, to facilitate replication of the experiments using Finite Element Modeling (FEM)

Balancing Mass

The balancing mass, given by “Balancing_Mass_kg” in the “Test Info.csv” file, was a counterbalance applied to each specimen in order to counteract the effect of gravity and ensure the specimens started from a fully neutral position. The mass used for each test was equivalent to the mass of the fixturing plus one half of the mass of the potted specimen with mounting plates attached.

While in practice a series of pulleys was used, for FEM purposes this can be simplified by applying the equivalent force equally between the two balancing brackets (BOM Part #26) on either side of the fixture. The load application point is at the very end of the rectangular slot feature, as per the image below:

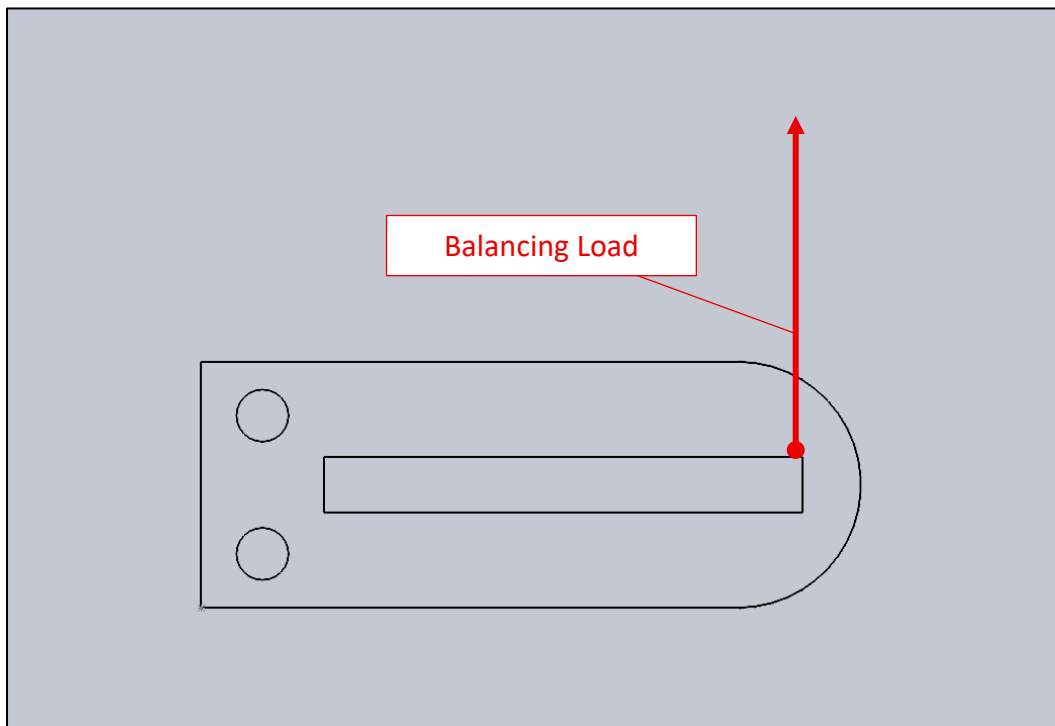


Figure 1 - Balancing load location on balancing bracket. Each bracket support one half of the total balancing mass.

Important: Please take care to only apply half of the balancing force to each of the two balancing arms.

Shear Application Location

Shear was applied by displacing the M10x1.5 mm Screw (BOM Part # 30) upwards. This was done using a cable which was fed through the center hole drilled into the screw. Therefore, the displacement should be placed **directly at the center of the top of this hole**. Note that the cable cannot resist any other translations or any rotations, so the fixture must be translationally and rotationally unconstrained at this location aside from the displacement direction.

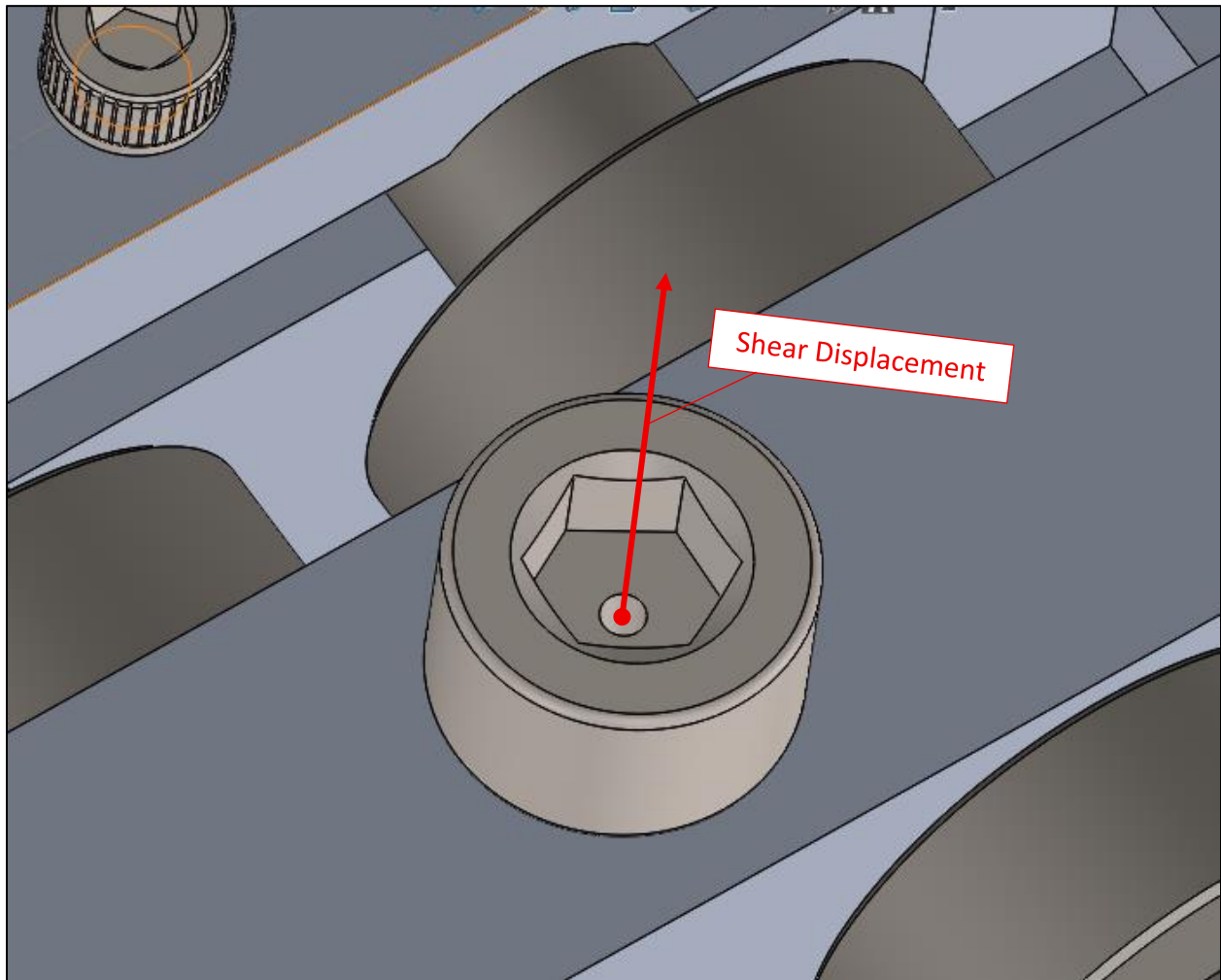


Figure 1 - Shear application location on carriage screw

Disc Offset

The disc offset (given by “Disc_Offset_mm” in the “Test Info.csv” file) is used to ensure that the shearing load is applied directly through the mid-disc plane, and that no combined bending moments are applied to the specimen.

The disc offset is measured along the craniocaudal axis from the bottom of the cranial potting to the centroid of the intervertebral disc. The Disc offset has been highlighted in the below cutout of the simplified fixture schematic:

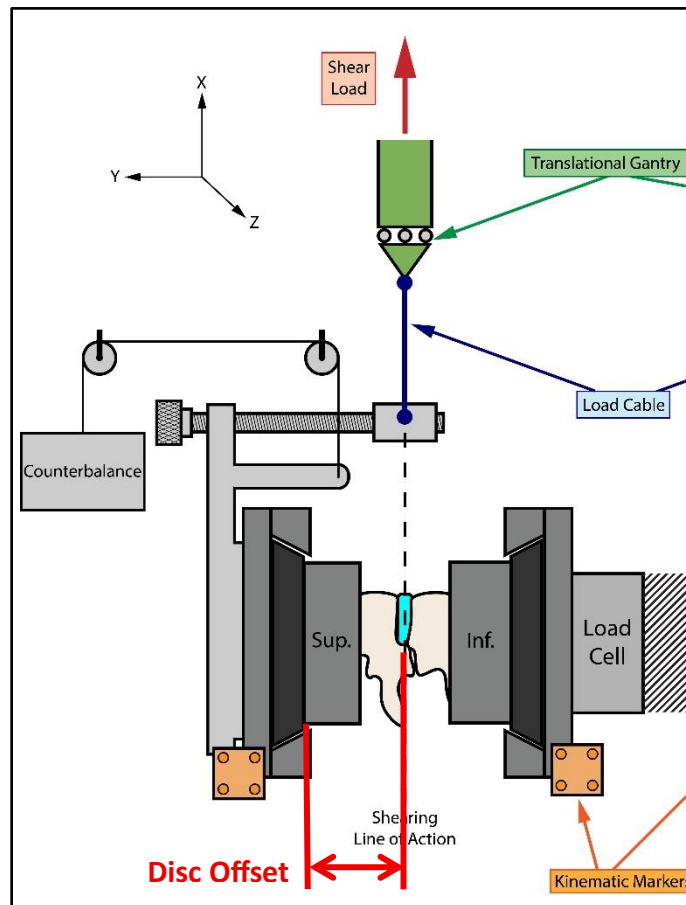


Figure 2 - Disc offset measurement on potted FSU in the shear loading fixture: between the bottom of the cranial/superior potting and the disc centroid.

The disc offset measurement should then be matched by the distance between the reference surfaces on the upper components of the fixture: namely, the distance (along the craniocaudal axis) between the reference surfaces on the Upper Inner Gantry Bracket (BOM Part #6) and the Gantry Block (BOM Part #3), as below:

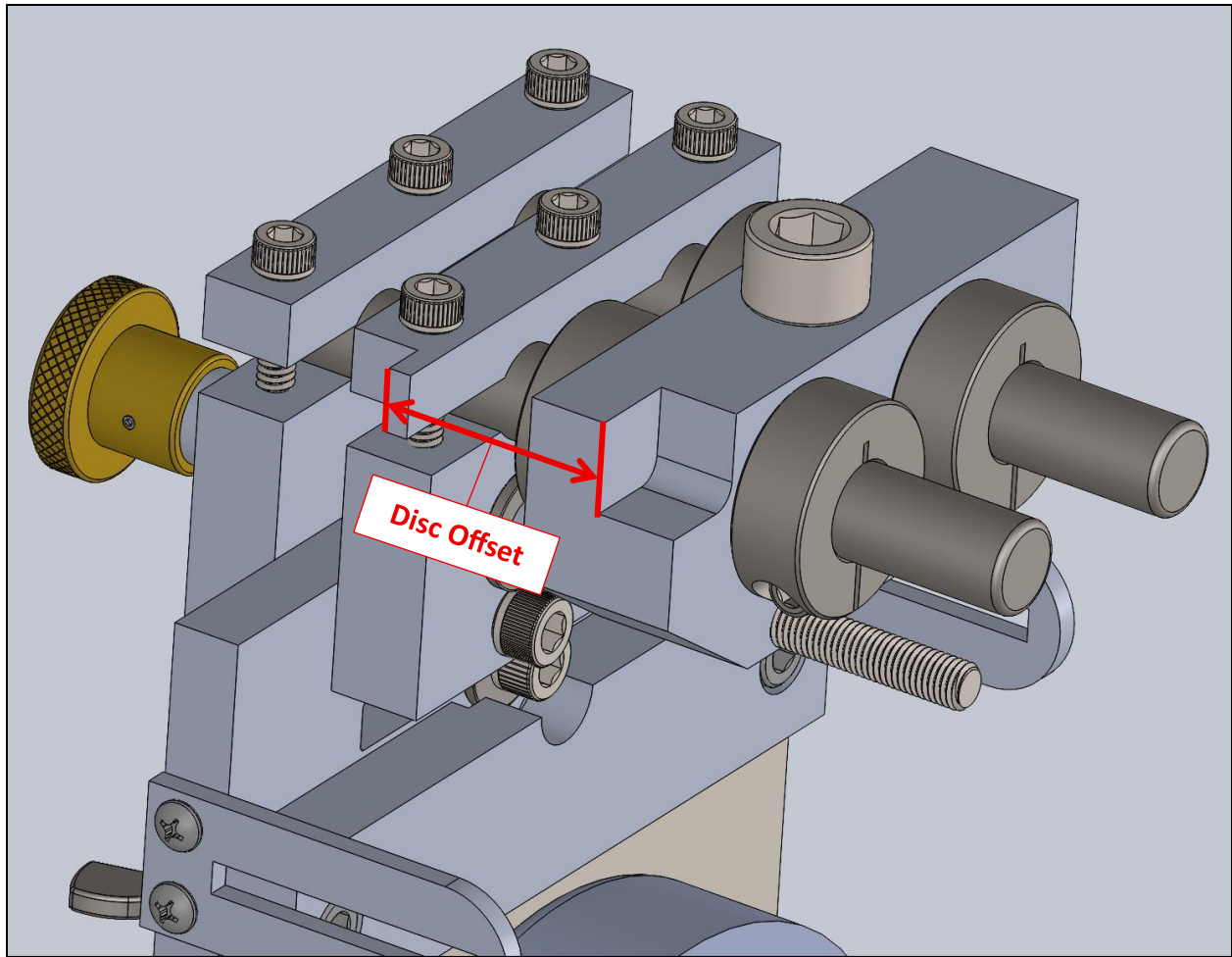


Figure 3 - Offset reference surfaces of the superior components of the fixture. The distance between the reference surfaces must be matched to the measured disc offset distance for each specimen.

A note on using these models in Finite Element Modeling (FEM) with Functional Spinal Units (FSUs):

Anatomical models of FSUs have not been included in this repository, and users must provide their own. When using a two-segment FSU model with this potting, the caudal FSU must be rigidly fixed in space, while the cranial FSU must be rigidly fixed to the cranial PMMA potting. The cranial vertebra can be attached to the potting at any depth so long as the intervertebral disc and facet joints are not occluded, and the loading line of action is appropriately aligned with the mid-disc plane using the Disc Offset measurement (see above). The specimen must be positioned with either the anterior or posterior direction facing up, depending on test orientation.

Care must be taken such that the intervertebral disc is coplanar with the shearing plane. This can be achieved using the average endplate angle as a landmark in the AP view, and the average angle of the anterior surfaces of the vertebral bodies as a landmark in the lateral view. A more detailed explanation of how this was achieved in our experiments can be found in the paper to which this repository belongs, in the methods sub-section titled “specimen preparation.”

Bill of Materials (BOM)

ITEM NO.	MCMMASTER-CARR PART NUMBER	DESCRIPTION	QTY.	Material
1	-	Backing Plate	1	6061-T6 Aluminum
2	5125K52	Brass Knob	1	Brass
3	-	Gantry Block	1	6061-T6 Aluminum
4	-	Gantry Rod	2	Stainless Steel
5	-	Inner Gantry Bracket, Lower	1	6061-T6 Aluminum
6	-	Inner Gantry Bracket, Upper	1	6061-T6 Aluminum
7	-	Outer Gantry Bracket, Lower	1	6061-T6 Aluminum
8	-	Outer Gantry Bracket, Upper	1	6061-T6 Aluminum
9	-	Spacer, Large	1	6061-T6 Aluminum
10	-	Spacer, small	1	6061-T6 Aluminum
11	92185A214	UNC #10-24 x1 1/4" SHCS	6	Stainless Steel
12	90499A029	UNC 1/4-20 Hex Nut	3	Steel
13	6435K34	1/2" Clamping Shaft Collar	4	Stainless Steel
14	90126A206	1/4" washer	4	Steel
15	92580A332	UNF 1/4-28 Adjustment Rod (cut to 5")	1	Steel
16	92185A544	UNC 1/4-20 x 1 1/4" SHCS	1	Stainless Steel
17	92185A548	UNC 1/4-20 x 1 3/4" SHCS	2	Stainless Steel
18	-	Lower Retainer	1	6061-T6 Aluminum
19	-	Upper Retainer	1	6061-T6 Aluminum
20	-	Specimen Mounting Plate	1	Stainless Steel
21	90866A029	Wing Nut	4	Steel
22	92196A540	UNC 1/4-20 x 3/4" SHCS	2	Stainless Steel
23	92196A544	UNC 1/4-20 x 1 1/4" SHCS	2	Stainless Steel
24	92311A581	UNC 5/16-18 x 3/4" Set Screw	2	Stainless Steel
25	91075A445	UNC #6-32 Standoff	2	Stainless Steel
26	-	Balancing Bracket	2	6061-T6 Aluminum
27	91772A146	UNC #6-32 x 3/8" Pan head screw	4	6061-T6 Aluminum
28	92196A537	UNC 1/4-20 x 1/2" SHCS	4	6061-T6 Aluminum
29	-	Cranial PMMA Potting	1	Bosworth Fastray Bone Cement
30	91292A046	M10 x 1.5 mm SHCS, 18 mm Long	1	Stainless Steel
31	92311A077	UNC #2-56 x 1/4" Set Screw	3	Stainless Steel

Notes:

- Components with no part numbers were manufactured in-house and are not available as off-the-shelf parts.
- All components with a part number were purchased from McMaster-Carr (<https://www.mcmaster.com/>).
- After purchase, component #2 was further modified with two tap holes and component # 15 was trimmed to size. Details on these modifications can be found in the model files.